Comparison to Related Bike-Sharing Demand Forecasting Studies

Metrics are not fully comparable because studies differ in city, aggregation level, target scale, and prediction horizon.

Work	Task / dataset	Best reported model	RMSE	MAE	R^2	MAPE	Main comparison point
Our project	**Capital Bikeshare, station-day, top 20 stations**	**Gradient Boosting + lag**	**22.78**	**16.92**	**0.663**	**25.3%**	**Stricter station-level setup; local noise remains visible**
Sathishkumar & Cho	Seoul Bike and Capital Bikeshare, system-level demand	CUBIST	n/a	n/a	0.95 / 0.89	n/a	Higher R^2, but system-level demand is easier than station-level demar
Sathishkumar et al.	Hourly Seoul Bike demand	Gradient Boosting	n/a	n/a	0.92	n/a	Hourly data includes strong time-of-day signal
Choi	Hourly Seoul Bike demand	Random Forest	282.63	169.57	0.77	n/a	Absolute errors are larger because target scale is larger
Xin	UCI-DC, one-hour-ahead forecasting	N-BEATS	75.88	51.33	n/a	26.7%	MAPE is close to our relative error level